9th Class • Punjab Board • PECTAA New Syllabus

Chapter 2: System Design and Troubleshooting

Complete solved exercise — digital logic, truth tables, Boolean simplification, troubleshooting and backups.

Chapter 2 • Complete official exercise • 9 MCQs + 10 short + 6 long questions

What this chapter is about

• **Digital systems** represent information using discrete values, especially binary 0 and 1. ADC converts an analog signal to digital form; DAC performs the reverse conversion.
• **Boolean algebra** uses AND, OR, NOT and related operations to describe digital logic. Truth tables list every input combination and its output.
• A **logic diagram** shows gates and their connections. Boolean laws allow an expression—and therefore its circuit—to be simplified.
• **Troubleshooting** is a systematic cycle: identify, form a theory, test, plan, implement, verify and document. Backups protect data if a repair or failure causes loss.

Nine-unit / twelve-unit coverage map

• This combined Chapter 2 covers **Unit 3 Digital Systems and Logic Design** and **Unit 4 System Troubleshooting** from the twelve-unit download.
• The main exercise follows the **April 2026 nine-unit PECTAA edition**. Section D clearly labels a small extension from the corresponding twelve-unit digital-logic unit.

A. Multiple Choice Questions

1. Which of the following Boolean expressions represents the OR operation?

(a) A · B
(b) A + B
(c) A
(d) A ⊕ B

Answer: (b) A + B — In Boolean notation, plus (+) denotes OR. A dot (·) denotes AND and ⊕ denotes XOR.

2. What is the dual of the Boolean expression A . 0 = 0?

(a) A + 1 = 1
(b) A + 0 = A
(c) A . 1 = A
(d) A . 0 = 0

Answer: (a) A + 1 = 1 — To form the dual, interchange AND with OR and 0 with 1: A · 0 = 0 becomes A + 1 = 1.

3. Which logic gate outputs true only if both inputs are true?

(a) OR gate
(b) AND gate
(c) XOR gate
(d) NOT gate

Answer: (b) AND gate — AND gives 1 only for the input combination A = 1 and B = 1.

4. What is the decimal equivalent of the binary number 1101?

(a) 11
(b) 12
(c) 13
(d) 14

Answer: (c) 13 — $1101_2 = 1\times8 + 1\times4 + 0\times2 + 1\times1 = 13_{10}$.

5. What is the first step in the systematic process of troubleshooting?

(a) Establish a Theory of Probable Cause
(b) Implement the Solution
(c) Identify Problem
(d) Document Findings, Actions, and Outcomes

Answer: (c) Identify Problem — First describe and confirm the exact problem; possible causes are considered afterwards.

6. Why is effective troubleshooting important for maintaining systems?

(a) It helps save money on repairs
(b) It prevents the need for professional help
(c) It ensures systems operate smoothly and efficiently
(d) It allows for more frequent system updates

Answer: (c) It ensures systems operate smoothly and efficiently — Its main purpose is reliable, smooth and efficient operation; lower repair cost is only an additional benefit.

7. Which step involves coming up with a theory about what might be causing a problem?

(a) Test the Theory to Determine the Cause
(b) Establish a Theory of Probable Cause
(c) Implement the Solution
(d) Verify Full System Functionality

Answer: (b) Establish a Theory of Probable Cause — This step proposes the most likely explanation before any test or repair is performed.

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8. After implementing a solution, what is the next step in the troubleshooting process?

(a) Document Findings, Actions, and Outcomes
(b) Test the Theory to Determine the Cause
(c) Verify Full System Functionality
(d) Establish a Plan of Action to Resolve the Problem

Answer: (c) Verify Full System Functionality — Verify that the original fault is gone and that the whole system works normally; document the work after verification.

9. Why is troubleshooting important in computing systems?

(a) It ensures hardware components are always up to date
(b) It prevents the need for data backups
(c) It helps keep systems running smoothly and securely
(d) It eliminates the need for software updates

Answer: (c) It helps keep systems running smoothly and securely — Troubleshooting restores reliable operation and can remove security problems; it never replaces updates or backups.

B. Short Questions

1. Define a Boolean function.

A Boolean function is a mathematical expression that combines binary variables with logical operations such as AND, OR and NOT to produce a binary output, 0 or 1. For example, F(A, B) = A · B is 1 only when both A and B are 1.

2. What is the significance of the truth table in digital logic?

A truth table lists every possible input combination and the corresponding output of a logic expression or circuit. It describes the circuit completely, helps verify Boolean simplifications and allows two expressions to be checked for equivalence.

3. Describe the function of a NOT gate with its truth table.

A NOT gate has one input and reverses it: 0 becomes 1 and 1 becomes 0. Its output is written A′ or $\bar{A}$.

NOT gate truth table

Input A	Output A′
0	1
1	0

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4. Write the purpose of ADC and DAC.

An **ADC** converts a continuous analog signal, such as microphone sound, into discrete digital values that a computer can process. A **DAC** converts digital values back into an analog signal, for example to drive speakers.

5. Make the truth of two variable A,B for AND operation.

For F = A · B, the output is 1 only when both inputs are 1.

Two-variable AND truth table

A	B	F = A · B
0	0	0
0	1	0
1	0	0
1	1	1

6. What is the first step in the systematic process of troubleshooting?

The first step is to **identify the problem**. Observe the symptoms, ask what changed, read error messages and describe the fault precisely—for example, “the laptop does not switch on even when the power button is pressed.”

7. After identifying a problem, what is the next step in troubleshooting?

Next, **establish a theory of probable cause**. Use the symptoms and prior knowledge to suggest the most likely reason, such as a loose power cable or a drained battery, so that it can be tested.

8. Describe the importance of testing a theory during the troubleshooting process.

Testing shows whether the suspected cause is correct before time or money is spent on a repair. A simple controlled test, such as trying a known-working power adapter, either confirms the theory or tells the technician to form another one.

9. Explain what the “Implement the Solution” step entails in a troubleshooting.

This step carries out the planned repair safely—for example, reconnecting a cable, replacing a faulty battery or reinstalling a driver. The technician follows the plan, protects data and avoids unrelated changes.

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10. Why,is it necessary to verify full system functionality after implementing a solution?

Verification confirms that the original fault has actually disappeared and that the repair has not caused another problem. The whole system should be tested under normal conditions, not just the single part that was repaired.

C. Long Questions

1. Explain the usage of Boolean operations in computers.

Boolean operations work on binary values and therefore match the two-state nature of digital circuits. They allow hardware to calculate, compare, select and control information.

• **Arithmetic and Logic Unit:** AND, OR, NOT and XOR combine with adder circuits to perform binary addition and other ALU operations.

• **Comparisons and decisions:** Conditions such as “both switches are ON” or “either request is valid” are evaluated as logical expressions and guide program or circuit decisions.

• **Data processing:** Bit masks use AND to clear selected bits, OR to set them and XOR to toggle them. These operations support flags, permissions and low-level data manipulation.

• **Control circuits:** Boolean functions determine when a motor, alarm, display or other component should turn on according to its input conditions.

• **Memory and digital devices:** Logic gates form adders, multiplexers, decoders, flip-flops and other building blocks used in processors and storage circuits.

• **Circuit simplification:** Boolean laws reduce an expression to fewer gates, making the final circuit cheaper, faster and more power-efficient while preserving the same truth table.

For example, F = A · B can represent an alarm that sounds only when sensor A **and** sensor B are active. F = A + B represents an alarm that sounds when at least one sensor is active.

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2. Simplify the following Boolean function using Boolean algebra rules: $F(A, B) = A \cdot B + A \cdot \bar{B}$

Here $\bar{B}$ means NOT B. Take A as the common factor, then use the complement and identity laws.

$$F(A, B) = A \cdot B + A \cdot \bar{B}$$

$$= A \cdot (B + \bar{B}) \text{ (Distributive law)}$$

$$= A \cdot 1 \text{ (Complement law: } B + \bar{B} = 1)$$

$$= A \text{ (Identity law: } A \cdot 1 = A)$$

Simplified function: F(A, B) = A

Truth-table check

A	B	A · B	A · $\bar{B}$	Original F	Simplified F = A
0	0	0	0	0	0
0	1	0	0	0	0
1	0	0	1	1	1
1	1	1	0	1	1

The output depends only on A, so the simplified circuit is just a connection from A to F; no AND, OR or NOT gate is required.

3. Discuss the importance of troubleshooting in maintaining the smooth operation of systems, especially computing systems.

Troubleshooting is important because a fault should be diagnosed by evidence rather than by guessing or replacing parts at random.

• **Prevents downtime:** Quick identification of a failed connection, service or component restores work and reduces lost productivity.

• **Protects data integrity:** Diagnosing storage errors, software faults and failing drives early can prevent corrupted or lost files. A verified backup adds further protection.

• **Improves security:** Removing malware, correcting unsafe settings and installing needed patches reduce opportunities for attack.

• **Reduces cost:** Testing a likely cause prevents unnecessary replacement of working hardware and avoids repeated repairs.

• **Extends equipment life:** Correcting overheating, dust, loose cables and storage problems prevents small faults from damaging other components.

• **Improves safety and reliability:** Power and hardware faults can be handled in a controlled way, while verification ensures that the repaired system works normally.

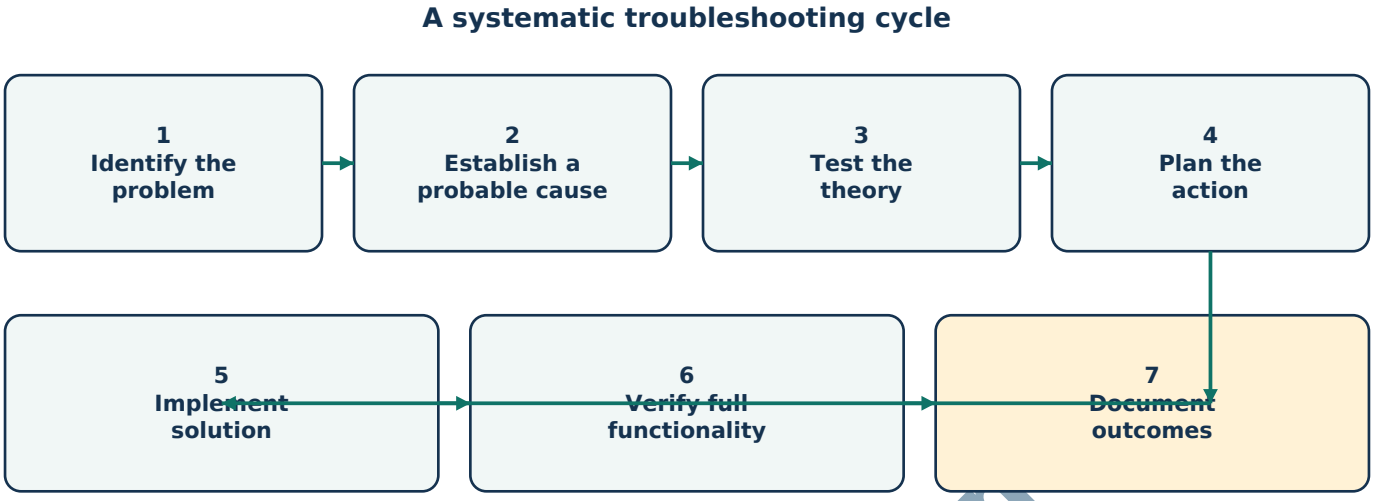
• **Builds knowledge:** Documented symptoms, tests and solutions make the same fault faster to solve next time and help other users.

Thus, troubleshooting is not only repair work; it is part of preventive maintenance and responsible system management.

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4. Explain the systematic process of troubleshooting. Describe each step in detail.

Figure 1 — Seven steps of systematic troubleshooting



1. Identify the problem: Gather symptoms, error messages and recent changes. Establish exactly what is not working and reproduce the fault safely if possible.

2. Establish a theory of probable cause: Suggest the most likely explanation, beginning with simple causes such as a disconnected cable or flat battery.

3. Test the theory to determine the cause: Perform a controlled check, swap in a known-good part or use a diagnostic tool. If the test disproves the theory, form and test another.

4. Establish a plan of action: Once the cause is confirmed, choose the safest and least disruptive repair. Consider data backup, required tools, cost and possible side effects.

5. Implement the solution: Carry out the plan carefully—for example, replace the battery, reconnect the device, remove malware or correct a setting.

6. Verify full system functionality: Repeat the original failing task and test related functions. Confirm normal performance and check that no new fault has appeared. If verification fails, revisit the theory.

7. Document findings, actions and outcomes: Record the symptoms, actual cause, tests, repair, replaced parts and final result for future reference.

5. Discuss the importance of documenting findings, actions, and outcomes during the troubleshooting process.

Documentation creates a reliable history of the fault and the work performed. It should record the date, user or device, symptoms, error messages, suspected causes, tests and their results, final cause, actions taken, parts changed and verification outcome.

• **Faster future repairs:** A technician can see what already worked or failed instead of repeating every test.

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• **Continuity:** Another technician can continue safely if the original person is unavailable.

• **Pattern recognition:** Repeated records may reveal that the same model, update or environment causes recurring faults.

• **Accountability:** Organisations can track replaced parts, costs, warranties and responsibility for configuration changes.

• **Learning and communication:** Clear notes become a troubleshooting guide and help explain the solution to the user.

• **Prevention:** Records of causes and outcomes support better maintenance, backup, training and purchasing decisions.

Documentation should be factual and should not expose passwords or sensitive data. “Replaced adapter; laptop now charges and runs a full battery test” is more useful than simply writing “fixed.”

6. Describe the different data backup methods.

A backup is a separate copy of data that can be restored after deletion, corruption, hardware failure, theft or malware. Methods can be classified by **what is copied** and **where the copy is stored**.

Backup type / medium	How it works	Advantage and limitation
Full backup	Copies all selected files every time.	Simplest restoration, but takes the most time and storage.
Incremental backup	Copies only changes made since the last backup of any type.	Fast and space-efficient; restoration needs the last full backup and every later incremental set.
Differential backup	Copies all changes made since the last full backup.	Restoration needs only the full and latest differential copies, but each later copy grows larger.
External storage	Copies data to an external HDD, SSD, USB drive or network storage.	Fast and under local control; the device can fail, be stolen or be affected by the same incident if always connected.
Cloud backup	Stores an encrypted copy on remote servers through the Internet.	Off-site protection and access from different places; requires Internet, account security and attention to privacy/cost.

A strong routine uses regular schedules, encryption where needed and a restore test. The **3–2–1 idea** is useful: keep three copies of important data, on two kinds of media, with one copy off-site or disconnected. Cloud synchronisation alone is not

always a backup because deletion or ransomware may also synchronise to the cloud.

D. More Important Questions and Official-Unit Extension

1. Differentiate between analog and digital signals.

Analog signal	Digital signal
Varies continuously over time and can take infinitely many values within a range.	Uses discrete levels; computer logic normally interprets them as 0 and 1.
Example: the original sound-wave voltage entering a microphone.	Example: sampled binary values stored in an audio file.
Noise can change the waveform directly and may be difficult to remove.	Error detection, regeneration and correction can make transmission and storage more reliable.
Converted to digital form by an ADC for computer processing.	Converted back to analog form by a DAC for devices such as speakers.

2. State important Boolean laws used for simplification.

Law	AND form	OR form
Identity	$A \cdot 1 = A$	$A + 0 = A$
Null / domination	$A \cdot 0 = 0$	$A + 1 = 1$
Idempotent	$A \cdot A = A$	$A + A = A$
Complement	$A \cdot \overline{A} = 0$	$A + \overline{A} = 1$
Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$
Associative	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$	$(A + B) + C = A + (B + C)$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + B \cdot C = (A + B)(A + C)$
De Morgan	$\overline{(A \cdot B)} = \overline{A} + \overline{B}$	$\overline{(A + B)} = \overline{A} \cdot \overline{B}$

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3. Twelve-unit extension: What are half-adders and full-adders, and why are K-maps useful?

Half-adder	Full-adder
Adds two one-bit inputs, A and B.	Adds A, B and an incoming carry C_{in}.
Sum = A ⊕ B	Sum = A ⊕ B ⊕ C_{in}
Carry = A · B	Carry-out = A·B + C_{in}·(A ⊕ B)
Used for the least significant bit or where no incoming carry exists.	Used for higher bits; full-adders can be chained to add multi-bit numbers.

A **Karnaugh map (K-map)** arranges a truth table so adjacent cells differ by only one variable. Groups of 1s containing 1, 2, 4, 8… cells reveal terms that can be combined. This gives a smaller Boolean expression and therefore fewer logic gates. This extension comes from the separately printed twelve-unit Digital Systems chapter; it is not an extra numbered exercise in the April 2026 combined edition.

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Points to Remember

ADC: analog → digital. **DAC:** digital → analog.

AND: output 1 only when all inputs are 1. **OR:** output 1 when at least one input is 1. **NOT:** reverses its one input.

Boolean dual: interchange AND/OR and 0/1. **Complement law:** $B + \overline{B} = 1$ and $B \cdot \overline{B} = 0$.

Truth table: lists all input combinations and outputs and proves whether two expressions are equivalent.

Troubleshooting order: identify → theory → test → plan → implement → verify → document.

Verification: test the original task and related functions; if the fault remains, return to a probable cause.

Documentation: record symptoms, tests, cause, actions, parts and final outcome without recording passwords.

Backups: full, incremental and differential describe what is copied; external and cloud describe where copies are kept.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	a	3	b	4	c	5	c
6	c	7	b	8	c	9	c		

How to attempt this chapter in the paper

- Draw truth tables with 2^n rows for n inputs and list combinations in binary order.
- In Boolean simplification, name the law used at every line and finish with a truth-table check where useful.
- For troubleshooting questions, keep the seven steps in order. Verification comes before final documentation.
- For backup answers, distinguish backup from ordinary synchronisation and mention restoration testing plus an off-site or disconnected copy.

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